

Highlights

Directional Transport and Geometric Confinement of Molecular Nanomachines on Bimetallic Width-Gradient Substrates

Keihan Javadi

- Continuous negative width gradients rectify the diffusive motion of diverse molecular nanomachines into directed, long-range transport.
- Geometric confinement on the NG surface minimizes angular velocity and maximizes effective velocity relative to zero- and positive-gradient substrates.
- Lightweight, flexible nanomachine designs excel at low temperatures, while structurally balanced, rigid designs dominate at high temperatures on the NG surface.

Directional Transport and Geometric Confinement of Molecular Nanomachines on Bimetallic Width-Gradient Substrates

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ABSTRACT

Molecular nanomachines hold immense potential for applications in targeted drug delivery, molecular transport, and nanoscale manipulation; however, achieving precise directional control over their diffusive motion remains a significant challenge. While previous research has extensively examined substrate topography and wheel composition, the specific influence of continuous width gradients on nanomachine mobility across varying thermal conditions is not fully understood. This study employs classical molecular dynamics (MD) simulations to investigate the performance of diverse nanomachines—including Nanocar, Miller nanocar, Nanotruck, Angled Nanocar, Nano-cooper, and Nano-caterpillar—on bimetallic substrates featuring a uniform gold path (Zero Gradient, ZG) alongside positive (PG) and negative (NG) width-gradient gold paths on a silver base, evaluated from 75 K to 600 K. The results reveal that although elevated temperatures generally enhance mobility by driving the dynamic transition from sliding to rolling, uniform and positive gradient surfaces exhibit delayed transitions and substantial lateral deviations. In contrast, the negative gradient surface significantly improves directional stability, minimizes angular velocity, and facilitates long-range displacement through synergistic geometric confinement and optimized surface energetics. Furthermore, while lightweight, flexible structures demonstrate superior mobility at lower temperatures, structurally balanced designs excel in high-temperature regimes on the NG surface. Ultimately, these findings establish that engineering negative width gradients is a highly effective strategy for achieving controlled, long-range, and thermally adaptive directional transport of molecular machines.

1. Introduction

The development of molecular machines has revolutionized nanotechnology, enabling precise manipulation and transport at the atomic scale [2, 4–7]. Inspired by natural biological motors, such as kinesin, synthetic molecular vehicles—commonly referred to as nanocars and nanotrucks—have been engineered with specialized chassis and molecular wheels to facilitate directed mobility across solid surfaces [1, 3]. These nanoscale devices hold immense potential for transformative applications, including targeted drug delivery, nanoscale fabrication, and molecular transport [4]. Among the various components of these molecular machines, the choice of wheels (e.g., *p*-carborane or fullerenes) and the chassis architecture (flexible versus rigid) play critical roles in determining their rolling efficiency, structural stability, and overall locomotion dynamics [1, 2].

A fundamental challenge in the practical deployment of molecular nanomachines is achieving precise directional control over their inherently diffusive, thermally driven motion [3]. On uniform flat substrates, nanomachines typically exhibit random, short-range fluctuations or long-range Brownian diffusion without a preferred direction [4]. To address this, recent computational and experimental studies have explored the use of complex substrate topographies, such as stepped edges, concave grooves, and conical geometries, to guide nanomachine trajectories and minimize lateral deviations [1, 3]. Furthermore, the influence of chassis flexibility and wheel size on the dynamic transition between

sliding and rolling regimes has been extensively documented [1, 2]. Despite these advancements, a significant research gap remains regarding the effect of continuous lateral width gradients on planar bimetallic substrates. While topographical height variations and radial gradients have been studied, the specific mechanism of geometric confinement induced by converging or diverging path widths—such as a gold track on a silver base—has not been systematically evaluated for its capacity to direct diverse nanomachine architectures across varying thermal conditions.

To bridge this gap, this study employs classical molecular dynamics (MD) simulations to investigate the thermally driven mobility of a comprehensive suite of nanomachines, including the Nanocar, Miller nanocar, Nanotruck, Angled Nanocar, Nano-cooper, and Nano-caterpillar. We introduce a novel bimetallic substrate configuration consisting of a gold path on a silver base, engineered with three distinct lateral width profiles: a uniform path (Zero Gradient, ZG), a diverging path (Positive Gradient, PG), and a converging path (Negative Gradient, NG). By evaluating the system across a broad temperature range (75 K to 600 K), this research aims to elucidate how width-induced geometric confinement and boundary energetics influence the dynamic transition from sliding to rolling, directional stability, and long-range displacement.

The remainder of this paper is organized as follows: Section 2 details the computational framework, force fields, and simulation setup. Section 3 presents the trajectory analyses, displacement metrics, and diffusion characteristics of the nanomachines on the gradient substrates. Section 3 interprets these findings in the context of geometric confinement and structural design, and Section 4 summarizes the key

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conclusions and their implications for future nanomachine engineering.

The mobility of molecular nanomachines on solid substrates has been the subject of extensive computational and experimental investigation. Early studies predominantly focused on flat gold surfaces, where nanomachines exhibit thermally activated diffusive motion characterized by short-range fluctuations at low temperatures and long-range Brownian diffusion at elevated temperatures [2, 4]. However, the absence of directional constraints on flat substrates results in significant lateral deviations, rendering them unsuitable for applications requiring controlled transport. To address this limitation, researchers have explored non-planar substrate geometries, including stepped surfaces and concave structures. Bakhtiari et al. [2] demonstrated that nanotrucks equipped with various fullerene wheels exhibit substantially more linear trajectories on downward-step substrates compared to flat or upward-step configurations, attributing this behavior to the energy barrier required for nanomachines to deviate from the predefined path. Similarly, Bakhtiari et al. [4] confirmed that the downward-step surface provides superior directional guidance for nanocars, with the lowest terrace emerging as the optimal surface in approximately 65% of tested conditions. These findings collectively establish that substrate topography plays a decisive role in regulating nanomachine trajectories.

Building upon the concept of topographical control, Shaygani et al. [3] introduced the novel concept of surface radius gradient (SRG) as a mechanism for directing nanomachine motion. By investigating the diffusive motion of C_{60} , nanocar, and nanotruck on conical and concave gold substrates with varying radius gradients, they demonstrated that negative radius gradient (NRG) surfaces consistently outperform positive radius gradient (PRG) configurations in promoting long-range, directional displacement. The NRG geometry reduces deviations by progressively guiding nanomachines away from high-energy edge regions, while simultaneously facilitating long-range movement even at low temperatures (75 K and 150 K). Furthermore, Bakhtiari et al. [1] showed that the concave substrate, which integrates features of both flat and stepped geometries, provides the most effective control over nanomachine movement by minimizing edge-induced deviations while maintaining the capacity for directed motion. These studies collectively highlight the potential of geometric confinement—whether through radial gradients or concave topographies—as a strategy for enhancing directional stability. However, the specific influence of continuous lateral width gradients on planar bimetallic substrates has not been systematically investigated, representing a gap that the present study aims to address.

The selection of molecular wheels constitutes another critical design parameter governing nanomachine mobility. Bakhtiari et al. [2] conducted the first systematic evaluation of nanotrucks equipped with different fullerene wheels (C_{36} , C_{50} , C_{60} , C_{76} , C_{80} , and C_{90}) on flat, upward-step, and downward-step gold substrates. Their results revealed

that the conventional C_{60} wheel is optimal in only 17% of conditions, with smaller fullerenes (C_{36} , C_{50}) demonstrating superior traveled distances at higher temperatures due to their lower inertia, while larger fullerenes (C_{80} , C_{90}) exhibit enhanced rolling behavior at lower temperatures owing to their more spherical geometry. Extending this analysis to nanocars, Bakhtiari et al. [4] confirmed that C_{60} -based nanocars are optimal in merely 10% of evaluated conditions, further challenging the universality of C_{60} as the default wheel choice. These findings underscore the importance of tailoring wheel size to the specific operational conditions, including substrate geometry and temperature range. Comparable sensitivity of fullerene-wheeled nanomachines to curved gold-substrate geometry has also been reported [9].

The structural architecture of the nanomachine chassis profoundly influences its locomotion characteristics. Bakhtiari et al. [1] compared the mobility of the flexible Miller nanocar and the rigid Nano-caterpillar across flat, stepped, and concave gold substrates at temperatures ranging from 75 K to 700 K. The Miller nanocar, with its flexible chassis, exhibited superior long-range mobility but experienced larger deviations at higher temperatures due to increased conformational degrees of freedom. In contrast, the Nano-caterpillar's rigid structure ensured greater directional stability but limited its overall range of motion. A similar trade-off was observed by Shaygani et al. [3], who noted that the flexible-chassis nanocar demonstrated enhanced adaptability to surface gradients but was more susceptible to lateral deviations, whereas the rigid-chassis nanotruck maintained more consistent trajectories across varying substrate conditions. These observations establish a fundamental design trade-off: flexibility enhances maneuverability at the cost of directional precision, while rigidity ensures stability but constrains the range of accessible motion.

Temperature emerges as a governing parameter across all investigated systems, modulating the dynamic transition between sliding and rolling regimes. At low temperatures (75–150 K), nanomachines are predominantly confined to short-range fluctuations and local oscillations due to insufficient kinetic energy to overcome inertial effects [2, 4]. As temperature increases, the kinetic energy surpasses the activation barrier, enabling long-range displacement and triggering the transition from sliding to rolling motion. Bakhtiari et al. [1] observed that this transition occurs at approximately 300 K on flat substrates, while Shaygani et al. [3] reported that surface gradients can modulate the onset temperature of this transition. The coexistence of sliding and rolling during the transitional regime amplifies trajectory oscillations and increases angular deviations, a phenomenon consistently reported across all four reference studies [1–4]. Understanding how substrate geometry and width gradients influence this transition remains an open question of considerable practical significance.

In summary, the existing literature has established the critical roles of substrate geometry, wheel configuration, chassis design, and temperature in governing nanomachine

mobility. However, several gaps persist. First, prior studies have predominantly employed monometallic gold substrates, whereas bimetallic configurations (e.g., gold paths on silver bases) may introduce additional energetic and geometric effects at material boundaries. Second, while radial gradients [3] and discrete steps [2, 4] have been investigated, continuous lateral width gradients—offering a distinct mechanism of geometric confinement—have not been explored. Third, the simultaneous evaluation of diverse nanomachine architectures (spanning flexible and rigid chassis designs with varying wheel arrangements) on gradient substrates across a broad thermal range has not been reported. The present study addresses these gaps by investigating six nanomachine designs on bimetallic substrates with uniform, positive, and negative width-gradient gold paths, providing a comprehensive assessment of geometric confinement effects on directional mobility, sliding-to-rolling transitions, and long-range transport efficiency.

2. Materials and method

Classical molecular dynamics (MD) simulations were employed to investigate the thermally driven mobility of molecular nanomachines using the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) software [11, 12]. The simulations were conducted in the canonical (NVT) ensemble to maintain precise temperature control across the system. A Nosé-Hoover thermostat was applied independently to the nanomachines and the floating layer of the substrate, with damping parameters of 0.050 ps and 0.005 ps, respectively [14]. The equations of motion were integrated using the velocity-Verlet algorithm with a time step of 1 fs (0.001 ps). Prior to the production runs, the systems underwent energy minimization to eliminate any initial steric clashes or high-energy configurations, followed by a 1 ns production run (10^6 steps) to ensure adequate sampling of the phase space. The visualizations and trajectory analyses were subsequently performed using Visual Molecular Dynamics (VMD) software [13].

Six distinct nanomachine architectures were simulated: the Nanocar, Miller nanocar, Nanotruck, Angled Nanocar, Nano-cooper, and Nano-caterpillar. These models encompass a variety of chassis flexibilities, masses, and wheel configurations to provide a comprehensive evaluation of structural effects on mobility. The substrate was modeled as a bimetallic system consisting of a gold (Au) track deposited on a silver (Ag) base. To assess the impact of geometric confinement and directional guidance, three distinct path geometries were engineered, characterized by boundary angles of 45° , 90° , and 135° . These configurations correspond to a Positive Gradient (PG) path, where the track width diverges; a Zero Gradient (ZG) uniform path with a constant width; and a Negative Gradient (NG) path, where the track width converges. The bottommost atomic layer of the substrate was kept rigid to simulate a bulk material foundation, while the intermediate layers were allowed to relax dynamically. The simulations were executed across a broad thermal range,

specifically at 75 K, 150 K, 300 K, 400 K, 500 K, and 600 K, to capture the transition from localized fluctuations to long-range diffusive and rolling regimes.

The interactions within the system were modeled using a hybrid potential approach. The metallic interactions between the gold and silver atoms in the substrate were described by the Embedded Atom Method (EAM) alloy potential, which accurately captures the many-body effects and cohesive energy of face-centered cubic (FCC) metals [2, 3]. The intramolecular interactions within the nanomachines, including bond stretching, angle bending, and dihedral torsions, were modeled using the Molecular Mechanics (MM) force field. The bond and angle potentials were represented by harmonic functions:

$$E_{\text{bond}} = K_b(r - r_0)^2 \quad (1)$$

$$E_{\text{angle}} = K_a(\theta - \theta_0)^2 \quad (2)$$

where K_b and K_a are the bond and angle stiffness constants, r and θ are the instantaneous bond distance and angle, and r_0 and θ_0 are their respective equilibrium values. The dihedral torsions were governed by the Optimized Potential for Liquid Simulations (OPLS) style:

$$E_{\text{dihedral}} = \frac{1}{2}V_1 [1 + \cos(\phi)] + \frac{1}{2}V_2 [1 - \cos(2\phi)] + \frac{1}{2}V_3 [1 + \cos(3\phi)] + \frac{1}{2}V_4 [1 - \cos(4\phi)] \quad (3)$$

where V_1 through V_4 are the torsional rigidity coefficients and ϕ is the dihedral angle.

The non-bonded van der Waals interactions between the nanomachine atoms (Carbon, Hydrogen, Nitrogen) and the substrate atoms (Gold, Silver), as well as intermolecular interactions, were modeled using the 12-6 Lennard-Jones (LJ) potential:

$$E_{\text{LJ}}(r) = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r} \right)^{12} - \left(\frac{\sigma_{ij}}{r} \right)^6 \right] \quad (4)$$

where ϵ_{ij} represents the depth of the potential well, σ_{ij} is the distance at which the intermolecular potential is zero, and r is the distance between atoms i and j . A cut-off radius of 13 Å was applied to all LJ interactions. The cross-interaction parameters for ϵ_{ij} and σ_{ij} were derived using arithmetic mixing rules, consistent with established protocols for organic-metal interfaces [1, 4]. Electrostatic interactions were computed using partial atomic charges, though charge transfer between the neutral gold substrate and the nanomachines was considered negligible.

To quantitatively evaluate the mobility and directional stability of the nanomachines, several kinematic metrics were extracted from the simulation trajectories. The Mean

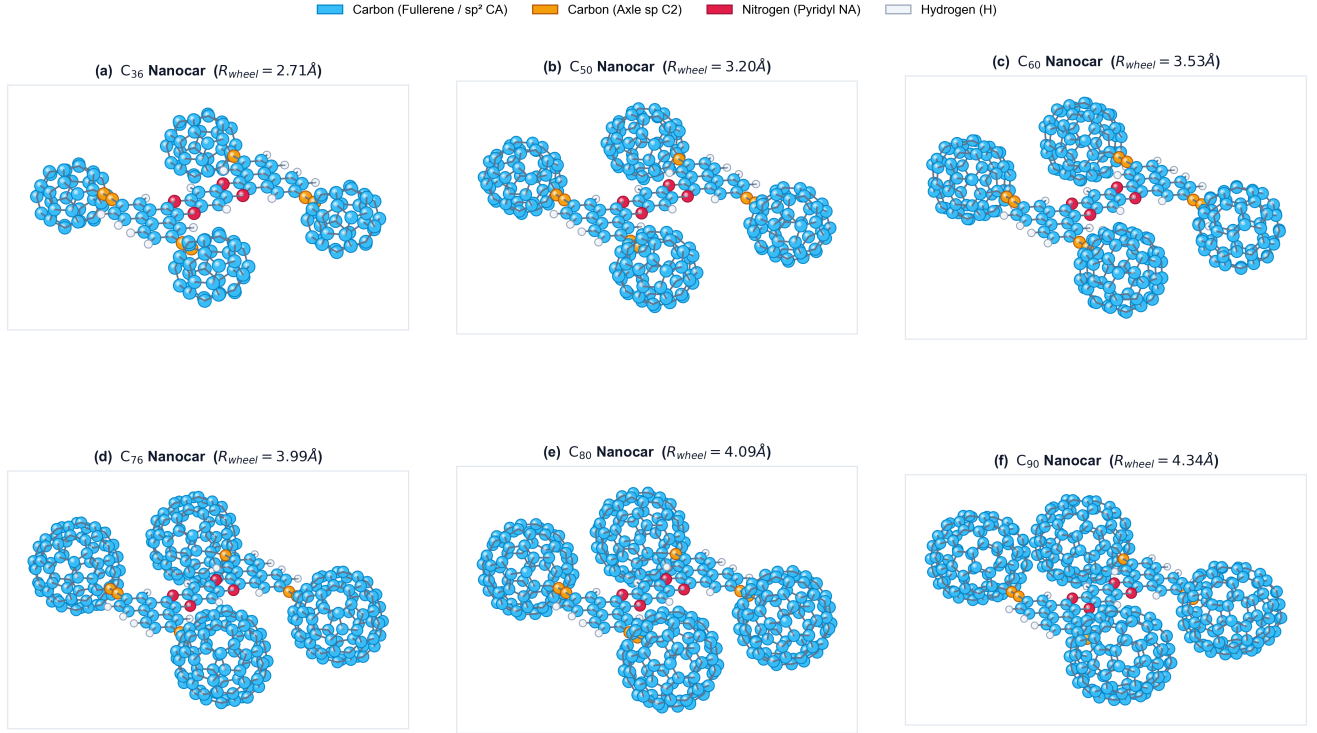


Figure 1: Different nanotrucks made of the selected fullerenes on the flat surface. A) a nanocar made of C₃₆ as the wheel. B) a nanocar made of C₅₀ as the wheel. C) a nanocar made of C₆₀ as the wheel. D) a nanocar made of C₇₆ as the wheel. E) a nanocar made of C₈₀ as the wheel. F) a nanocar made of C₉₀ as the wheel.

Squared Displacement (MSD) was computed to characterize the diffusive regime of the nanomachines:

$$\text{MSD}(t) = \langle |\mathbf{r}(t) - \mathbf{r}(0)|^2 \rangle \quad (5)$$

where $\mathbf{r}(t)$ and $\mathbf{r}(0)$ denote the center-of-mass position of the nanomachine at time t and the initial position, respectively. The diffusion coefficient (D) was subsequently derived from the linear regime of the MSD curve using the Einstein relation:

$$D = \lim_{t \rightarrow \infty} \frac{\text{MSD}(t)}{2dt} \quad (6)$$

where d represents the dimensionality of the surface diffusion. Furthermore, to assess the quality of directional transport, the *effective velocity* (V_{eff}) and *angular velocity* (ω) were calculated. The effective velocity isolates the translational kinetic contribution along the primary axis of the gradient (the X-axis), defined as the ratio of the velocity component in the desired direction to the total magnitude of the velocity vector. Conversely, the angular velocity quantifies the rate of rotational deviation from the primary trajectory, serving as a direct measure of directional stability and geometric confinement efficacy.

3. Results & discussion

3.1. Reference Surface: Uniform Gold Path

To evaluate the intrinsic performance of the nanomachines before the application of any gradient, a uniform gold

path with constant lateral dimensions of $40 \times 40 \text{\AA}$ on a silver substrate was simulated. This configuration has been established in previous studies as a suitable platform for nanomachine motion and was used in the present work as the reference surface for comparison with the gradient surfaces. The trajectories of the different nanomachines on this surface are presented in Figure 3.

At low temperatures, specifically 75 and 150 K, the kinetic energy of the nanomachines is insufficient to overcome inertial effects. Consequently, their motion is limited to short-range displacements and local oscillations. In this temperature range, lower-mass nanomachines with relatively flexible chassis structures, including the Nanocar, exhibit greater mobility than the other designs. In contrast, the Miller nanocar, despite its lower mass and flexible chassis, undergoes local oscillations and repeated changes in the direction of motion because its wheels are arranged in different orientations and no preferred direction of motion exists. As a result, this nanomachine does not convert its low mass and structural flexibility into effective displacement.

With increasing temperature, the kinetic energy of the nanomachines overcomes inertial effects and enables long-range displacement even for heavier structures. In this temperature range, all nanomachines enter the dynamic transition from the sliding regime to the rolling regime, consistent with previous studies [2, 4]. The coexistence of these two motion regimes intensifies trajectory oscillations and causes

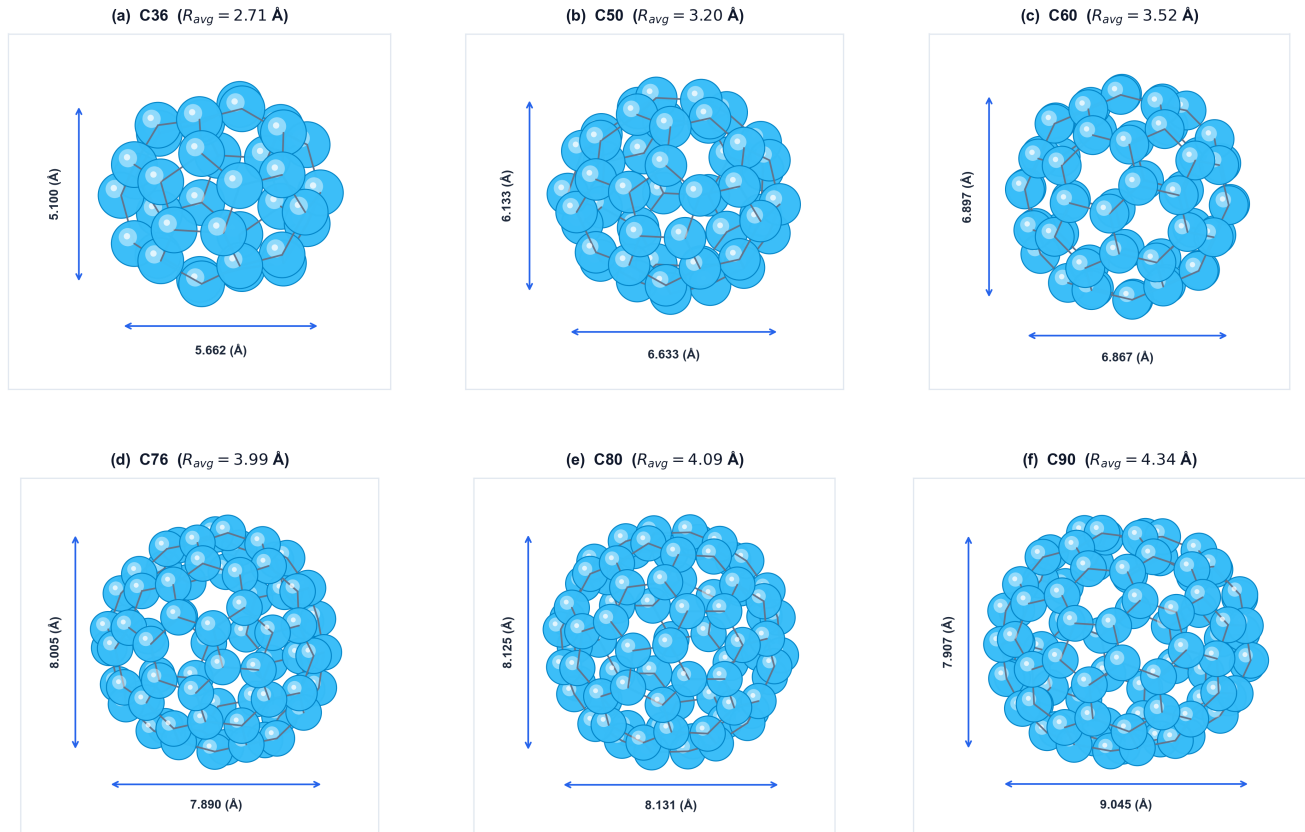


Figure 2: Different fullerenes used in the simulation with their radius. A) C36 with 2.71 Angstrom average radius. B) C50 with a 3.20 Angstrom average radius. C) C60 with a 3.54 Angstrom average radius. D) C76 with a 3.99 Angstrom average radius. E) C80 with 4.09 Angstrom average radius. F) C90 with 4.34 Angstrom average radius

all nanomachines to experience oscillatory motion and repeated changes in direction. This effect is observed as the increased tortuosity of the traveled paths in Figure 3.

At higher temperatures, namely 500 and 600 K, all nanomachines move entirely in the rolling regime and therefore experience their largest displacements on the surface. Nevertheless, the uniformity of the gold path and the absence of any structural or energetic gradient eliminate directional guidance. Under these conditions, the increase in traveled path length does not necessarily produce a larger net displacement in a specific direction. Among the examined nanomachines, the Nanotruck at 500 K and the Nanocar and Angled Nanocar at 600 K record the largest displacements.

The results obtained from the reference surface without a gradient demonstrate that increasing temperature enhances nanomachine mobility, while the uniform gold path prevents effective directional displacement. Therefore, to examine the effect of a geometric gradient on the motion behavior of the nanomachines, two gold paths with variable widths were designed and simulated. In the positive-gradient surface, denoted PG, the path width increases from 40 to 60 Å. In the negative-gradient surface, denoted NG, the path width decreases from 60 to 40 Å. The trajectories of the different nanomachines on these two surfaces are presented in Figures 4 and 5.

3.2. Motion on Positive- and Negative-Gradient Surfaces

At low temperatures, specifically 75 and 150 K, the nanomachines on both gradient surfaces mainly exhibit short-range displacements and oscillatory motion because their kinetic energy is insufficient to overcome inertial effects. However, the motion behavior on the two surfaces differs substantially. On the PG surface, the positive gradient regularizes the trajectory and reduces deviation from the main path relative to the uniform surface. The larger deviation of the Nanocar at 75 K is caused by its interaction with high-energy regions located at the gold–silver boundary. In contrast, the NG surface provides more favorable conditions for motion: more than 80% of the nanomachines experience longer-range displacements even at these low temperatures.

With increasing temperature, the dynamic transition of the nanomachines from the **sliding** regime to the **rolling** regime is delayed on both gradient surfaces relative to the reference surface. This transition occurs at 400 K instead of 300 K. On the PG surface, the delay persists up to 500 K, and a substantial fraction of the nanomachines retains sliding-dominated behavior. On the NG surface, however, long-range motion is continuously facilitated at intermediate temperatures. The Miller nanocar and Nano-cooper at 300 K,

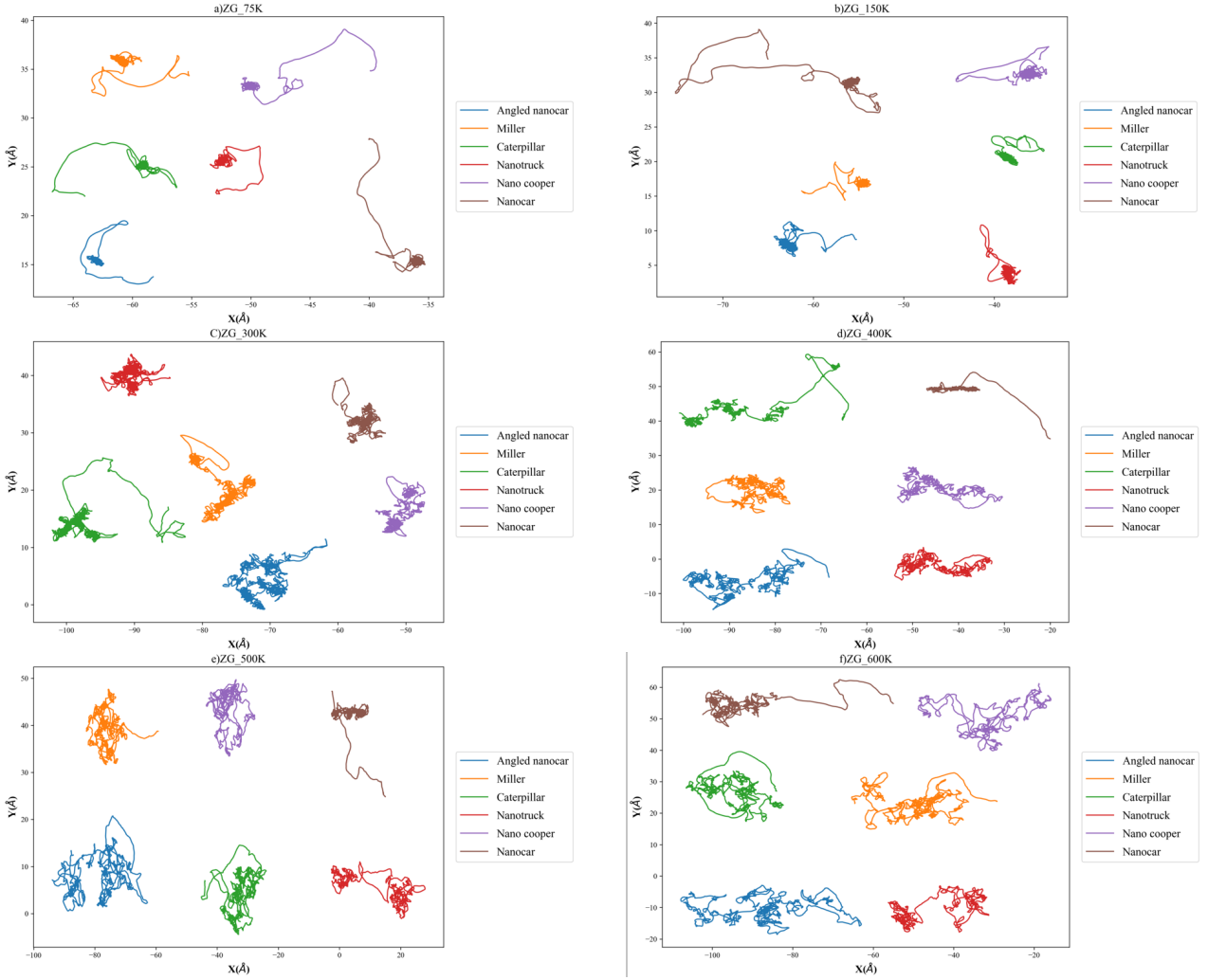


Figure 3: Zero Gradient Trajectories

and the Angled Nanocar and Nanotruck at 400 K, exhibit the largest displacements among the examined nanomachines.

At high temperatures, namely 500 and 600 K, the performance difference between the two gradient surfaces becomes more pronounced. All nanomachines on the NG surface experience substantial displacements, whereas on the PG surface a substantial fraction of the nanomachines, including the Miller nanocar and Nano-caterpillar, remains dominated by oscillatory motion even at 600 K. The trajectory comparison demonstrates that the negative gradient substantially facilitates directional motion and long-range displacement. The positive gradient regularizes the trajectory to some extent but is less effective than the negative gradient in enhancing nanomachine mobility, consistent with prior findings that substrate gradients and curvature can strongly influence nanoscale trajectories [3, 9].

3.3. Traveled Distance, Displacement, and Diffusion Coefficient

Figure 6 shows the distance traveled by the different nanomachines on the ZG, PG, and NG surfaces at the examined temperatures. In most temperatures, the nanomachines travel a greater distance on the ZG and NG surfaces than on the PG surface. This behavior agrees with the trajectory analysis and demonstrates that the delayed sliding-to-rolling transition on the PG surface restricts long-range motion and reduces the traveled distance. The traveled distance on the NG surface also exceeds that on the ZG surface in most temperatures. This observation is consistent with the trajectories shown in Figures 4 and 5 and confirms the role of the negative gradient in facilitating nanomachine motion.

On all surfaces, lighter nanomachines, particularly the Miller nanocar and the Nanocar, travel greater distances than the other structures in most temperatures. This behavior results from their lower mass and more flexible chassis relative to those of Nano-cooper, Nanotruck, and Angled Nanocar. Although heavier nanomachines generally travel

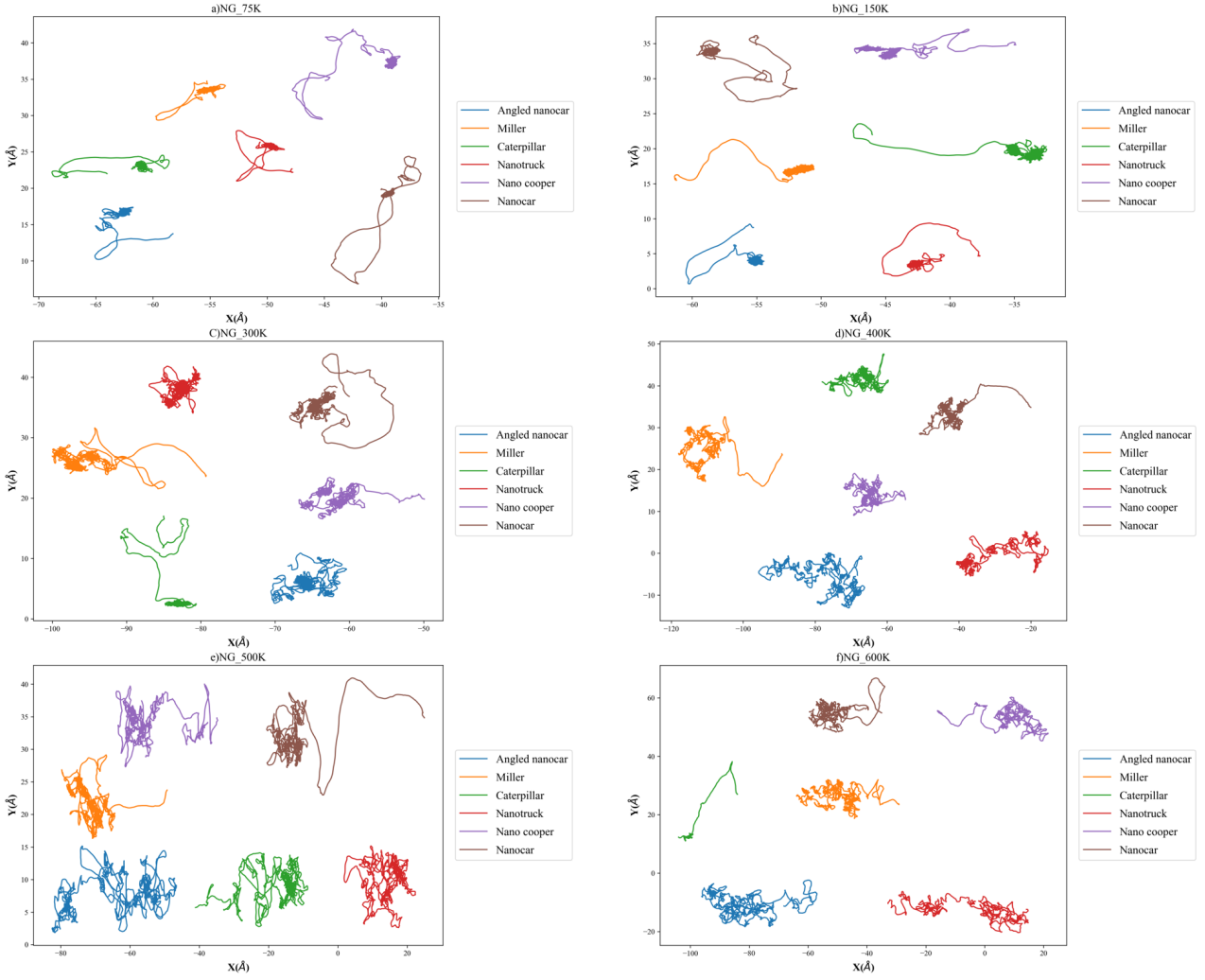


Figure 4: Negative Gradient Trajectories

shorter distances because of their larger mass and higher inertia, the Nano-caterpillar performs better within this group. Its larger number of wheels compensates for the limitations associated with higher mass and inertia and enables longer traveled distances.

Figure 7 presents the average displacement of the different nanomachines on the ZG, PG, and NG surfaces. This quantity is one of the most important metrics for evaluating nanomachine performance. In most temperatures, the highest average displacement belongs to the NG surface, followed by the ZG surface, while the PG surface shows the lowest value. These results are fully consistent with the trajectory and traveled-distance analyses. The only notable exception is the Angled Nanocar on the PG surface at 500 K, where it records a larger displacement than the other nanomachines on that surface.

On all three surfaces, the average displacement increases with temperature, agreeing with prior reports of thermally activated nanocar and nanotruck mobility [2–4, 7]. At low temperatures, limited kinetic energy and the dominance of

oscillatory and short-range motions keep the net displacement small. As temperature increases, kinetic energy overcomes inertial effects, the nanomachines undergo longer-range motion, and the contribution of local oscillations to their trajectories decreases. Consequently, the net displacement increases substantially. This trend is consistent with the trajectory and traveled-distance results and agrees with previous reports [2–4].

In addition to temperature, displacement depends on the intrinsic structural characteristics of the nanomachines. The governing factors include mass, inertia, chassis rigidity, wheel number, and wheel arrangement. The Miller nanocar, despite its lower mass and inertia, is limited in maintaining motion along a specific direction and achieving effective displacement because of its limited wheel number, wheel arrangement, and high chassis flexibility. The Nano-caterpillar, by contrast, has a higher capacity for straight motion because of its larger number of wheels and their appropriate arrangement, although its higher mass and inertia limit its performance at low temperatures. The Nanocar, Nanotruck, and Angled Nanocar possess a more balanced

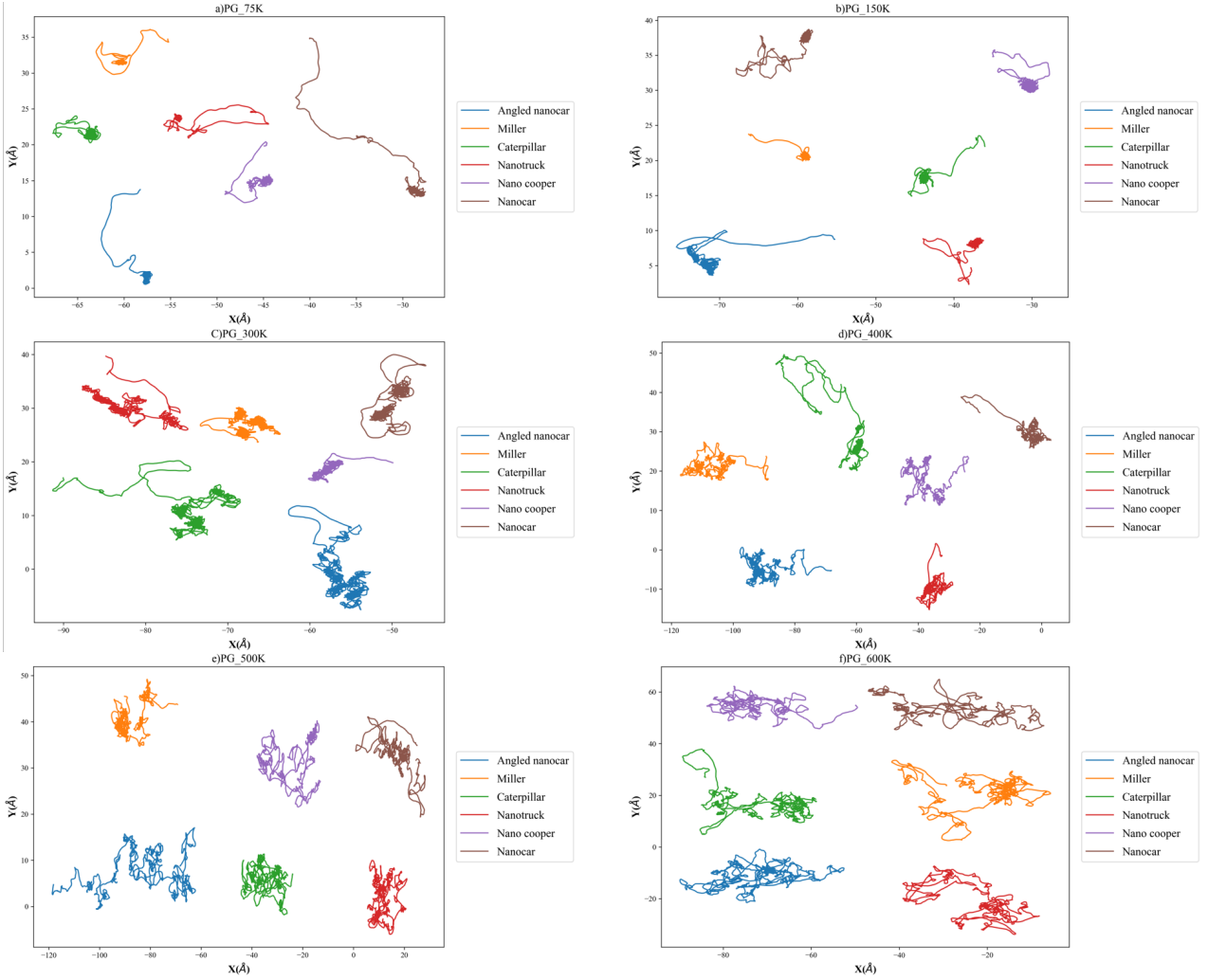


Figure 5: Positive Gradient Trajectories

combination of mass, chassis rigidity, and wheel arrangement and therefore exhibit the largest displacements in most conditions.

Figure 8 shows the diffusion coefficients of the different nanomachines on the three examined surfaces. These coefficients are fully consistent with the trends observed in the trajectories, traveled distance, and displacement. In most temperatures, heavier nanomachines exhibit larger diffusion coefficients than lighter structures. This result demonstrates that although higher mass increases inertia and restricts motion, the structural characteristics of these nanomachines, particularly their larger wheel number and appropriate wheel arrangement, compensate for this restriction and increase surface mobility.

Furthermore, the diffusion coefficients exceed $0.01 \text{ Å}^2 \text{ ps}^{-1}$ in the examined conditions. Based on previous studies [4], values above this threshold indicate that the nanomachines have left the short-range motion regime and entered the diffusive regime. This result is consistent with the increases in traveled distance and displacement reported above.

3.4. Velocity and Directional Stability

Figure 9 presents the average velocity of the different nanomachines on the ZG, PG, and NG surfaces at the examined temperatures. The velocity trend is fully consistent with the traveled-distance results shown in Figure 6 and demonstrates that nanomachine mobility increases substantially with temperature. At low temperatures, specifically 75 and 150 K, the kinetic energy is limited, motion is dominated by local oscillations and short-range displacements, and the velocity differences among the structures are negligible.

As temperature increases, the role of structural characteristics becomes more pronounced. Lighter nanomachines reach higher velocities than heavier structures because of their lower mass and inertia; this difference becomes larger at higher temperatures. The PG surface exhibits the lowest average velocity among the three surfaces in most temperatures. This behavior results from the delayed transition from the sliding regime to the rolling regime, which reduces mobility and restricts long-range motion. The NG surface facilitates this transition and provides more favorable conditions for higher velocities.

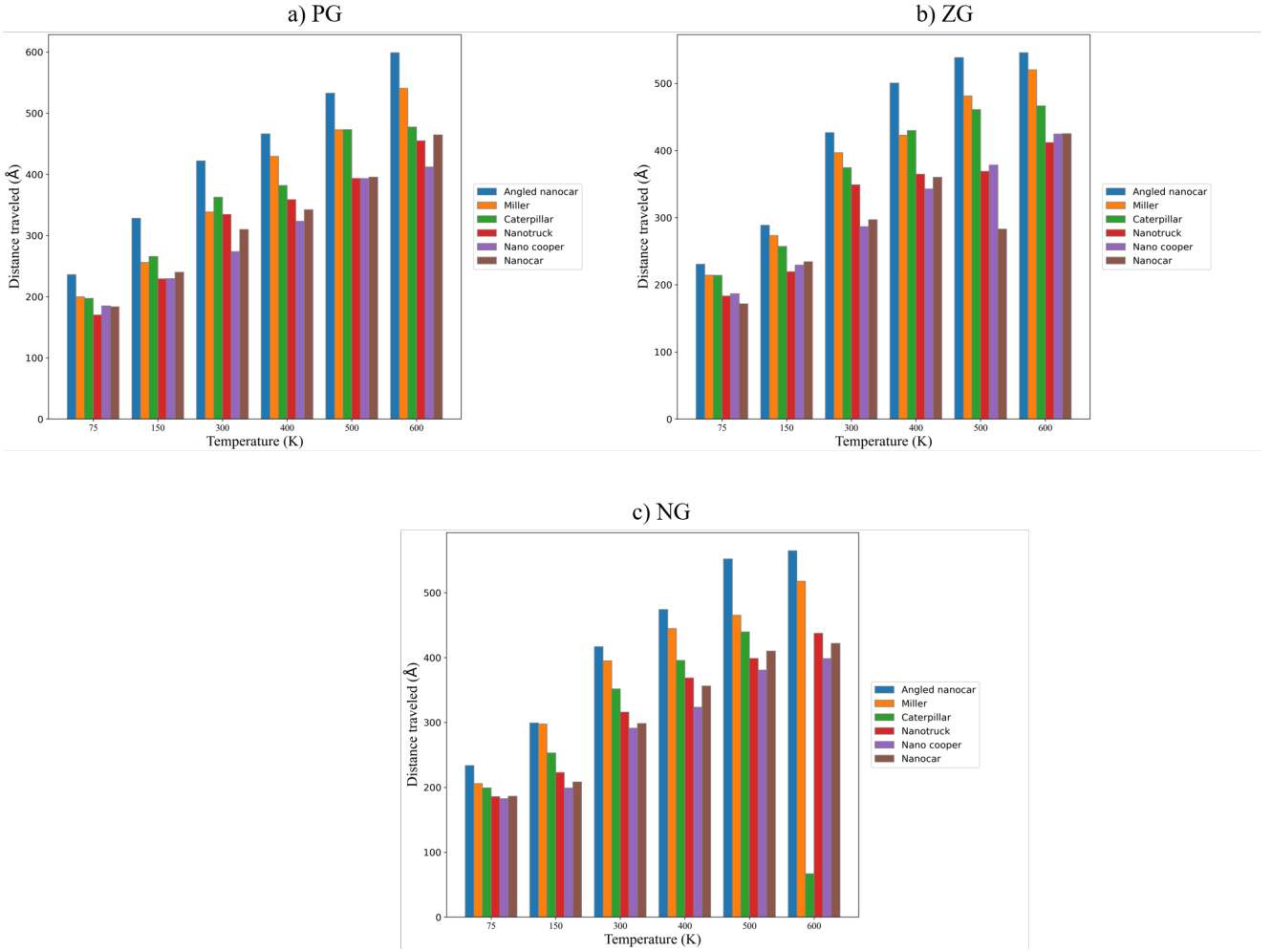


Figure 6: Nanocar Traveled Distance on Different Gradients

Although increased velocity and displacement indicate improved mobility, these metrics alone do not describe the quality of directional motion. Therefore, two additional metrics, **effective velocity** and **angular velocity**, were evaluated to assess the ability of the nanomachines to maintain a straight path. Effective velocity is the fraction of the velocity component along the desired direction of motion, namely the x -axis, relative to the total velocity. Angular velocity quantifies the deviation and rotation of the nanomachine from this direction.

Figure 10 shows the effective velocity of the different nanomachines on the ZG, PG, and NG surfaces. Unlike the average velocity, which increases substantially with temperature, the effective velocity exhibits limited variation and no clear dependence on nanomachine type or temperature. This behavior demonstrates that increasing thermal energy amplifies lateral oscillations almost proportionally to the increase in translational speed; consequently, the fraction of motion along the desired direction remains essentially constant. The PG surface records the lowest effective velocity in most

temperatures. This result is caused by the delayed sliding-to-rolling transition, which prolongs lateral oscillations and reduces directional-motion efficiency. The NG surface exhibits the highest effective velocity, demonstrating the positive role of the negative gradient in maintaining motion along the main axis.

To identify the origin of this difference, Figure 11 compares the angular velocity of the nanomachines on the three surfaces. Because this parameter measures changes in the direction of motion, lower values indicate greater path stability and more effective guidance. Among the examined structures, the Nanocar exhibits the highest angular velocity. This behavior results from its higher chassis flexibility and the associated increase in rotational degrees of freedom, which enhance sensitivity to thermal fluctuations and facilitate repeated changes in direction. In contrast, the Miller nanocar, despite its low mass, exhibits a relatively low angular velocity. This behavior results from its symmetric wheel arrangement and the absence of a dominant direction of motion; most of its motion remains confined to local oscillations and does not produce large angular rotations.

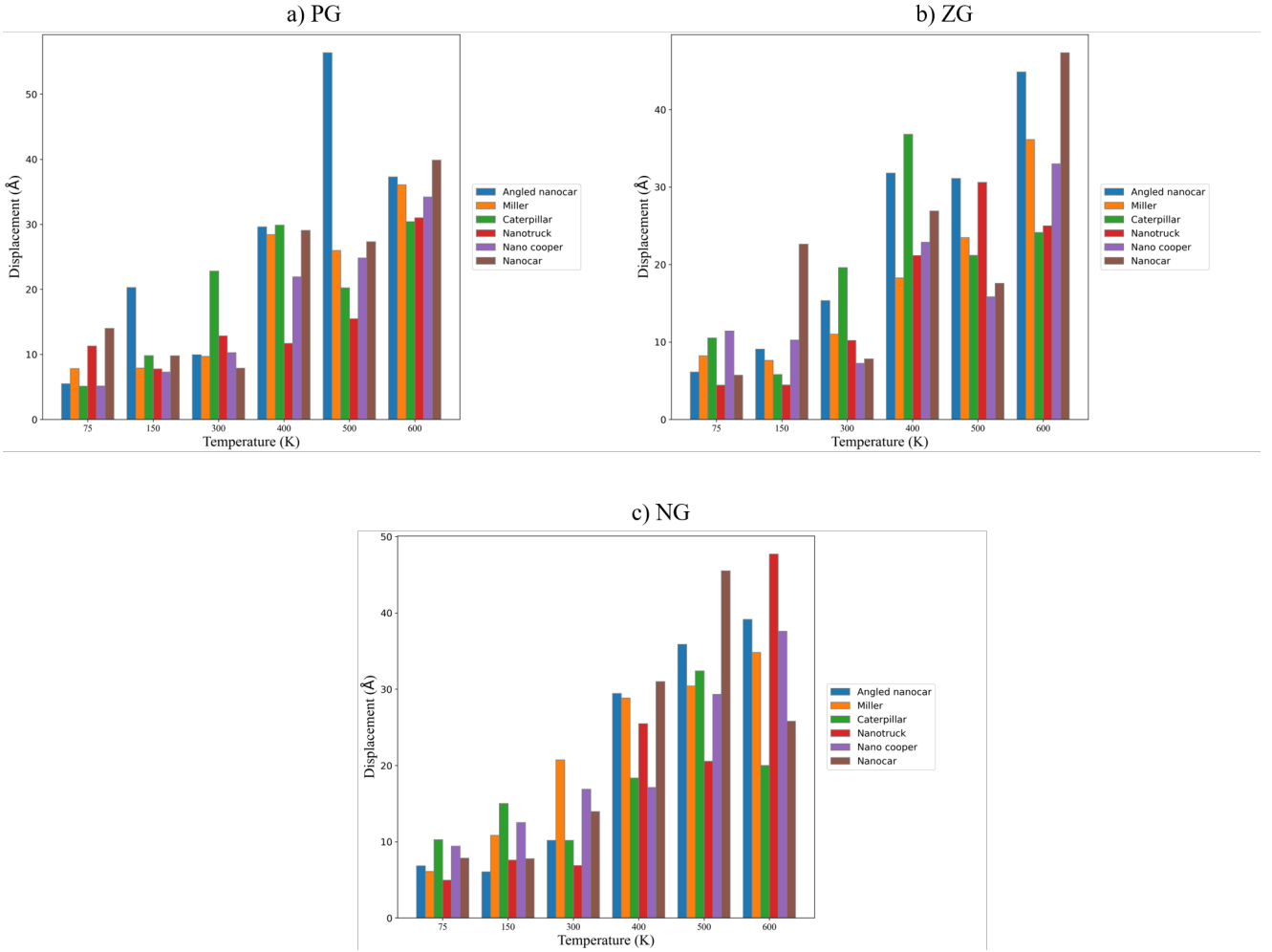


Figure 7: Nanocar Displacement on Different Gradients

The comparison of the three surfaces shows that the NG surface produces the lowest angular velocity at all temperatures, corresponding to the highest directional stability. This behavior results from two concurrent mechanisms. First, the negative gradient facilitates the transition from sliding to rolling and thereby enables more stable nanomachine motion. Second, the gradual reduction of the gold-path width along the direction of motion continuously restricts lateral degrees of freedom and reduces deviation from the main path. As a result, a larger portion of the kinetic energy is devoted to translational motion along the gradient, and the contribution of lateral oscillations decreases. This synergy between geometric confinement and surface-energy characteristics parallels observations that substrate shape and imposed gradients can steer molecular-machine motion [3, 9, 10] and explains the superiority of the NG surface in maintaining directional motion relative to the ZG and PG surfaces.

3.5. Optimal Nanomachines and Optimal Surfaces

Table 1 lists the best-performing nanomachines on each of the ZG, PG, and NG surfaces at the different temperatures.

The results demonstrate that the superiority of any design is not independent of environmental conditions. The optimal performance of each nanomachine arises from the interaction between its structural characteristics and the conditions imposed by the surface and temperature.

At low temperatures, specifically 75 and 150 K, lighter nanomachines, particularly the Nanocar and Angled Nanocar, perform better on the ZG and PG surfaces. In this temperature range, limited kinetic energy makes lower mass and inertia decisive for initiating motion. On the NG surface, however, the Nano-caterpillar is the best-performing nanomachine at both low temperatures. This result demonstrates that the negative gradient compensates for part of the limitation associated with higher mass by providing more stable conditions for motion.

With increasing temperature, the performance pattern changes, and different designs become superior on each surface. This behavior demonstrates that at higher temperatures nanomachine performance is not governed solely by mass; chassis rigidity, wheel number and arrangement, and the interaction with the surface also become decisive. Under these

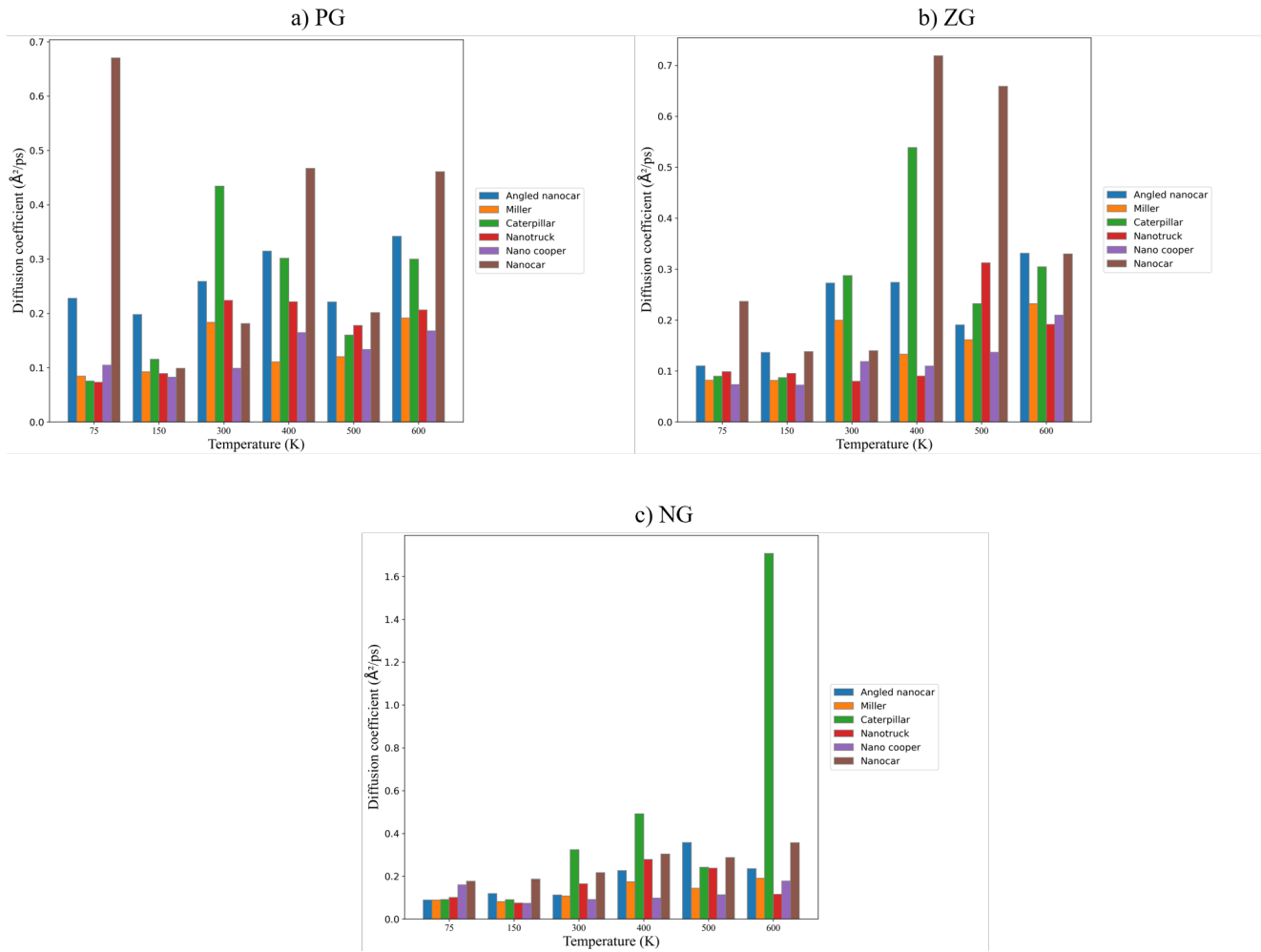


Figure 8: Nanocars Diffusion Coefficient

	75 K	150 K	300 K	400 K	500 K	600 K
ZG	Angled nanocar	Nanocar	Caterpillar	Caterpillar	Nanotruck	Nanocooper
PG	Nanocar	Angled nanocar	Caterpillar	Miller	Angled nanocar	Nanocar & Nanocooper
NG	Caterpillar	Caterpillar	Miller & Nanocooper	Nanotruck	Nanocar	Nanotruck

Table 1

Best-performing nanomachine for each substrate and temperature.

conditions, the Nanotruck and Nano-cooper are the best-performing nanomachines in several cases, demonstrating the increased importance of structural stability in long-range motion.

Table 2 identifies the best surface for each nanomachine at the different temperatures. The results demonstrate that no surface has absolute superiority under all conditions for all nanomachines. The effectiveness of each surface depends on the structural characteristics of the nanomachine and the ambient temperature. Nevertheless, a general trend is observable in the data.

At low temperatures, specifically 75 and 150 K, surface selection depends strongly on the nanomachine design. For example, the Angled Nanocar and Nanotruck achieve their highest performance mainly on the PG surface, whereas the Nanocar and Miller nanocar generally perform better on the ZG and NG surfaces in this temperature range. This behavior demonstrates that, at low temperatures where kinetic energy is limited, the interaction between surface geometry and the intrinsic characteristics of each nanomachine governs performance.

With increasing temperature, the share of the NG surface increases substantially and becomes the dominant surface

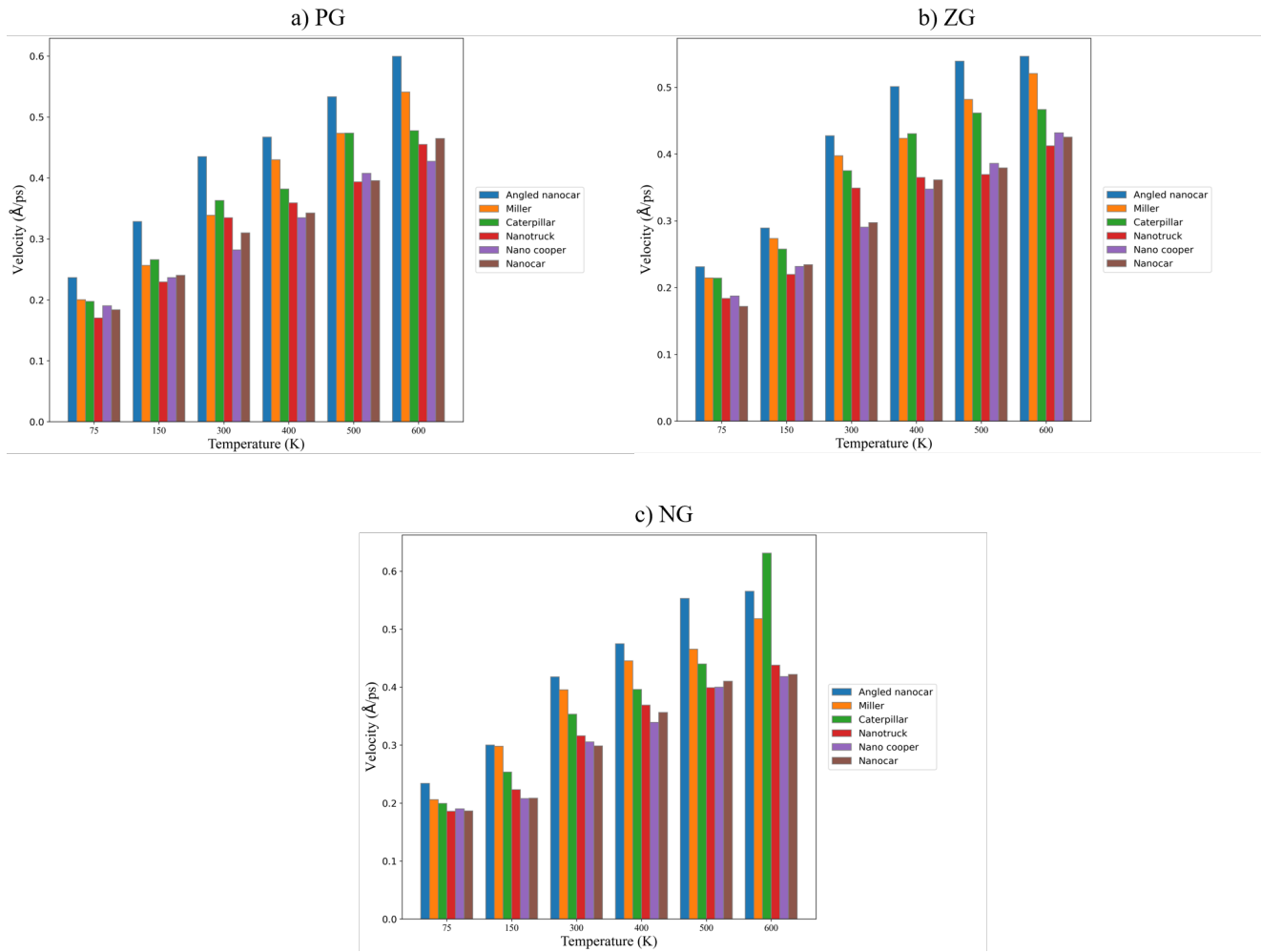


Figure 9: Nanocar Velocities

	75 K	150 K	300 K	400 K	500 K	600 K
Angled Nanocar	PG	PG	ZG	ZG	PG	ZG
Miller	ZG	NG	NG	PG	NG	ZG
Caterpillar	NG	NG	PG	ZG	NG	PG
Nanotruck	PG	NG	PG	NG	NG	NG
Nanocooper	NG	NG	NG	ZG	NG	PG
Nanocar	ZG	ZG	NG	ZG	NG	PG

Table 2

Optimal substrate for each nanomachine at different temperatures.

for most nanomachines. The Nanotruck at 400, 500, and 600 K, Nano-cooper at 300–600 K, and the Nano-caterpillar at 150, 300, and 500 K record their highest performance on the NG surface. These results demonstrate that the negative gradient, by increasing kinetic energy and facilitating directional motion, more effectively reveals the intrinsic capabilities of a wide range of nanomachines.

In contrast, the PG surface is the best surface only in a limited number of conditions, and its superiority is confined

mainly to low temperatures or specific designs. This behavior is consistent with the previous results and demonstrates that the delayed sliding-to-rolling transition on this surface reduces its efficiency at higher temperatures. Overall, Table 2 demonstrates that the NG surface has the greatest compatibility with the different nanomachine designs and provides the best conditions for directional and stable motion over the widest temperature range.

The results of this study elucidate the complex interplay between thermal energy, substrate geometry, and

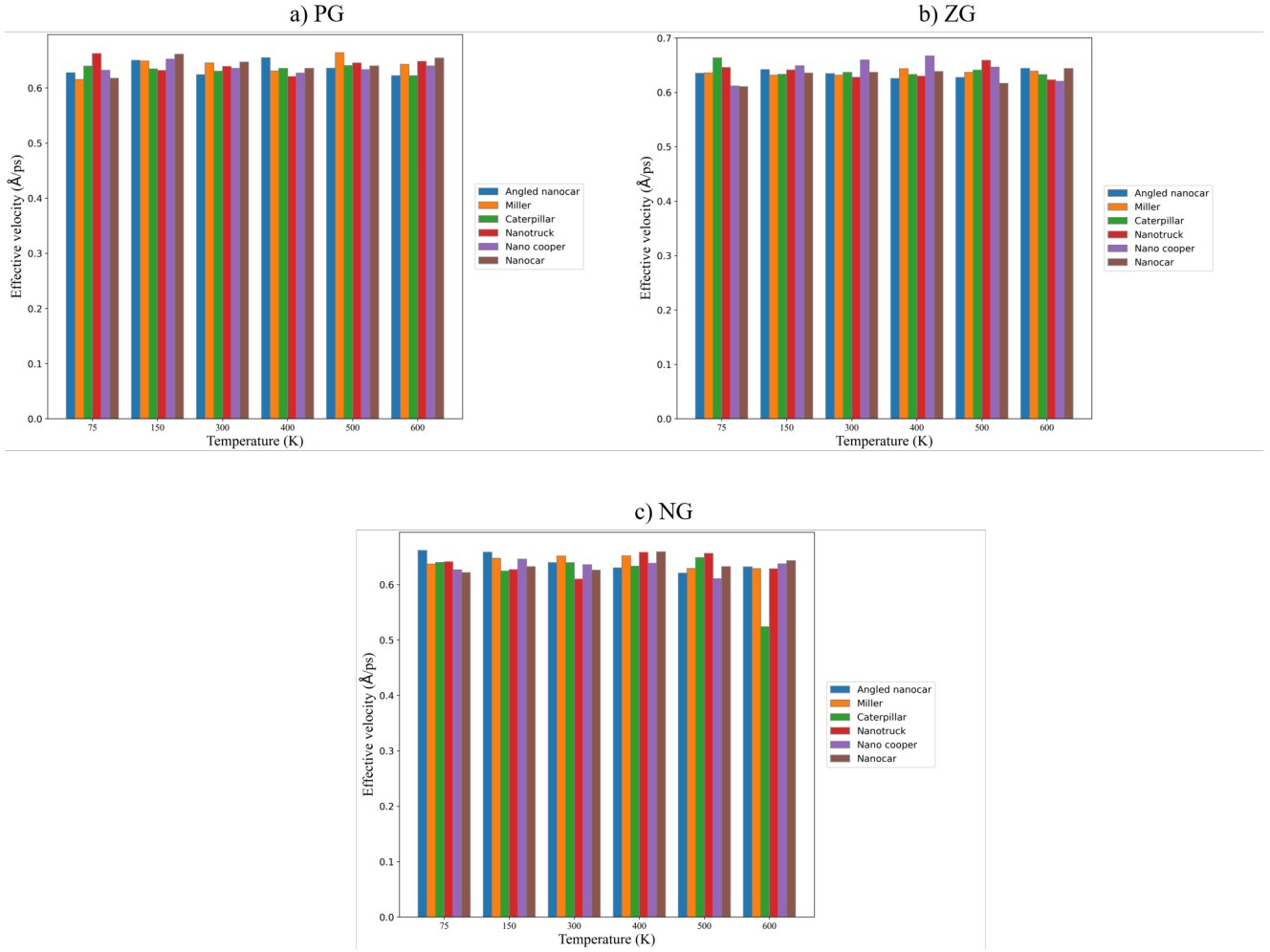


Figure 10: Effective Velocities

nanomachine architecture in governing directional transport at the molecular scale. A central finding is the temperature-dependent dynamic transition from the sliding regime to the rolling regime, which dictates the onset of long-range mobility. On the uniform Zero Gradient (ZG) surface, this transition occurs at approximately 300 K, consistent with previous observations for nanomachines on flat gold terraces [1, 4]. However, the introduction of lateral width gradients significantly modulates this transition. The Positive Gradient (PG) surface delays the onset of rolling to higher temperatures (up to 500 K), trapping the nanomachines in an oscillatory, sliding-dominated regime that severely restricts net displacement. Conversely, the Negative Gradient (NG) surface facilitates long-range motion at intermediate temperatures (300–400 K) by continuously guiding the nanomachines toward the center of the track, thereby promoting a more stable rolling transition. At elevated temperatures (500–600 K), while all nanomachines achieve full rolling, the uniform and diverging paths fail to provide directional guidance, resulting in high angular velocities and tortuous trajectories.

The superior performance of the Negative Gradient (NG) surface can be attributed to the mechanism of geometric

confinement. As the gold path converges, the lateral degrees of freedom of the nanomachines are progressively restricted. This funneling effect not only reduces the angular velocity but also maximizes the effective velocity by aligning the translational kinetic energy with the primary axis of motion. This phenomenon strongly parallels the findings of Shaygani et al. [3], who demonstrated that negative surface radius gradients (NRG) on conical and concave substrates minimize deviations by guiding nanomachines away from high-energy edge regions. Similarly, the guiding effect of the NG surface mirrors the directional control observed on downward-step substrates, where an energy barrier prevents nanomachines from easily deviating from the predefined path [2, 4], as well as steering effects on strained graphene substrates [10]. In contrast, the PG surface exposes the nanomachines to diverging boundaries; the interaction with the high-energy gold–silver interface at the widening edges induces lateral deviations, particularly for lightweight, flexible structures like the Nanocar at low temperatures.

The structural characteristics of the nanomachines—specifically mass, chassis rigidity, and wheel arrangement—exhibit a highly condition-dependent influence on mobility.

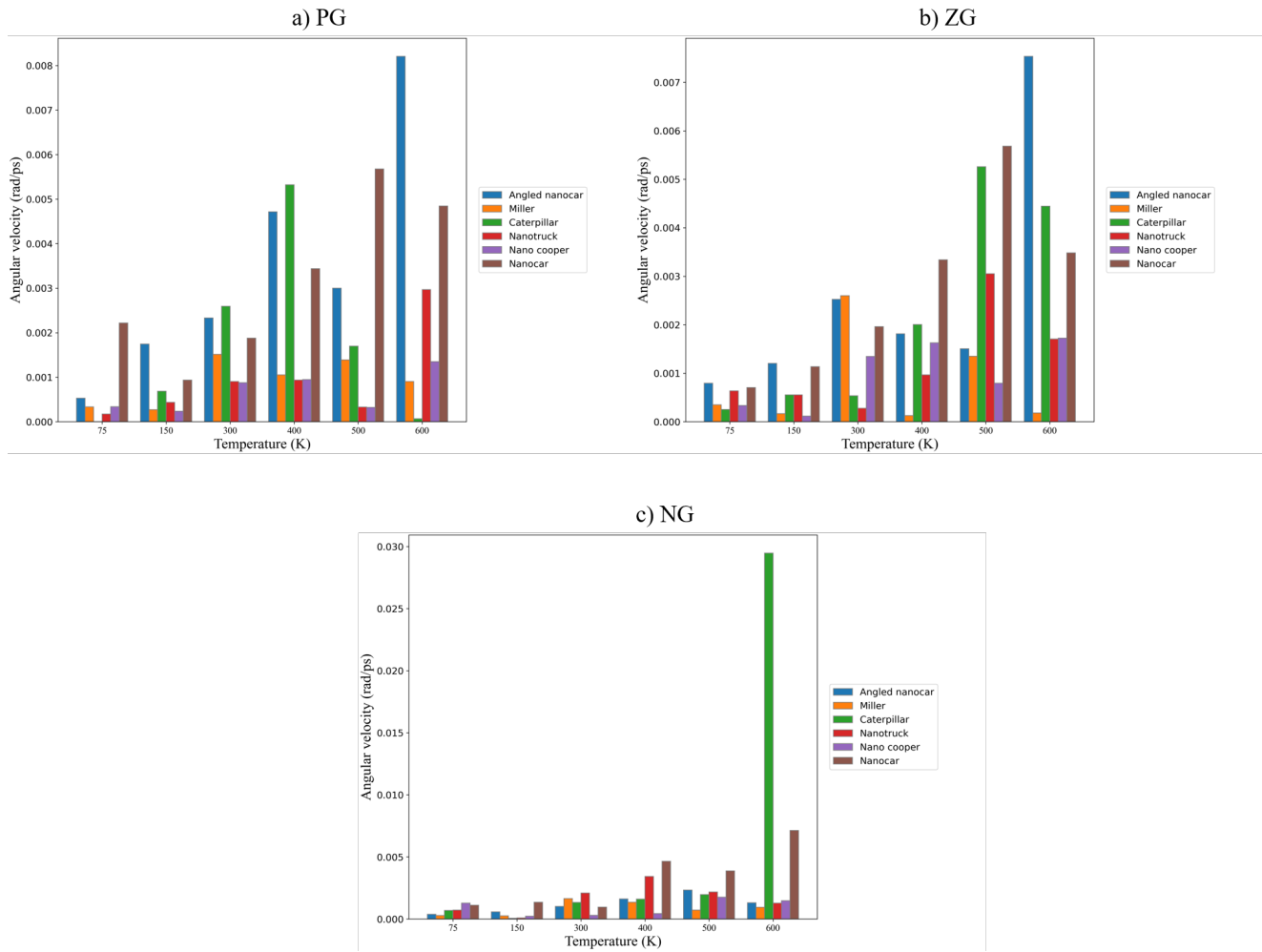


Figure 11: Nanocars Angular Velocities

At low temperatures (75–150 K), where kinetic energy is limited, lightweight and flexible designs such as the Nanocar and Miller nanocar overcome inertial barriers more readily, exhibiting superior short-range mobility on the ZG and PG surfaces. However, this flexibility comes at the cost of directional stability. The high angular velocity observed for the Nanocar across all surfaces confirms that a flexible chassis increases rotational degrees of freedom, amplifying sensitivity to thermal fluctuations and causing frequent directional changes, a trade-off previously identified by Bakhtiari et al. [1]. Conversely, heavier, more rigid structures like the Nanocaterpillar and Nanotruck suffer from higher inertia at low temperatures but demonstrate exceptional directional stability and long-range transport efficiency at elevated temperatures, particularly on the NG surface. The Nano-caterpillar, despite its high mass, leverages its multiple wheels to maintain continuous surface contact and stability, validating that structural balance often supersedes simple mass reduction in optimizing directed molecular transport [4].

Despite the comprehensive insights provided by this study, certain limitations must be acknowledged. First, the simulations rely on classical empirical potentials (EAM and

Lennard-Jones), which do not capture quantum mechanical effects such as dynamic charge transfer or polarization at the bimetallic gold–silver interface. While the assumption of a neutral substrate is standard and generally valid for these systems [2, 3], it may overlook subtle electrostatic steering effects. Second, the timescale of the simulations (1 ns) is sufficient to capture the fundamental diffusive regimes and dynamic transitions but may not fully represent the ultra-long-term macroscopic diffusion behavior or rare trapping events. Finally, the substrate models assume perfect crystalline structures without atomic defects, surface roughness, or adsorbates, which are inevitably present in experimental environments and could introduce localized energy traps that alter nanomachine trajectories.

The demonstration that continuous negative width gradients can effectively rectify the Brownian motion of diverse molecular machines into directed, long-range transport has profound implications for nanotechnology. These findings provide a foundational blueprint for the design of molecular assembly lines, nanoscale sorting devices, and targeted drug delivery vectors where precise spatial navigation is required. Future computational and experimental work should focus

on integrating external driving forces, such as electric or magnetic fields, with geometric gradients to achieve active, tunable control over nanomachine velocity and direction [8, 10]. Additionally, investigating the transport of cargo-loaded nanomachines on these bimetallic gradient tracks will be critical for evaluating the practical viability of these systems in real-world nanomedical and nanomanufacturing applications.

4. Conclusion

This study systematically investigated the thermally driven mobility of diverse molecular nanomachines on bimetallic substrates featuring uniform, positive, and negative width-gradient gold paths. By employing classical molecular dynamics simulations across a broad thermal range (75 K to 600 K), we demonstrated that while elevated temperatures generally enhance mobility by inducing a transition from sliding to rolling, the geometric confinement imposed by the substrate topology fundamentally dictates directional stability. The uniform and positive gradient surfaces exhibited delayed dynamic transitions and substantial lateral deviations, limiting their utility for controlled transport. In stark contrast, the negative gradient surface emerged as the optimal configuration, effectively funneling the nanomachines, minimizing angular velocity, and maximizing effective velocity through a synergy of geometric confinement and optimized boundary energetics. Furthermore, the analysis revealed a critical interplay between nanomachine architecture and thermal energy: while lightweight, flexible designs such as the Nanocar and Miller nanocar exhibit superior mobility at low temperatures, structurally balanced and rigid configurations, notably the Nanotruck and Nano-caterpillar, achieve unparalleled long-range directional stability at elevated temperatures on the converging track. Ultimately, these findings establish that engineering continuous negative width gradients is a robust and highly effective strategy for rectifying the Brownian motion of molecular machines into precise, long-range transport. This work provides a foundational blueprint for the design of next-generation nanoscale assembly lines, molecular sorting devices, and targeted drug delivery vectors, where reliable spatial navigation across complex environments is paramount.

CRedit authorship contribution statement

Keihan Javadi: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft.

Declaration of Competing Interest

There is no conflict of interest for this manuscript.

Data Availability

Data will be made available on request.

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